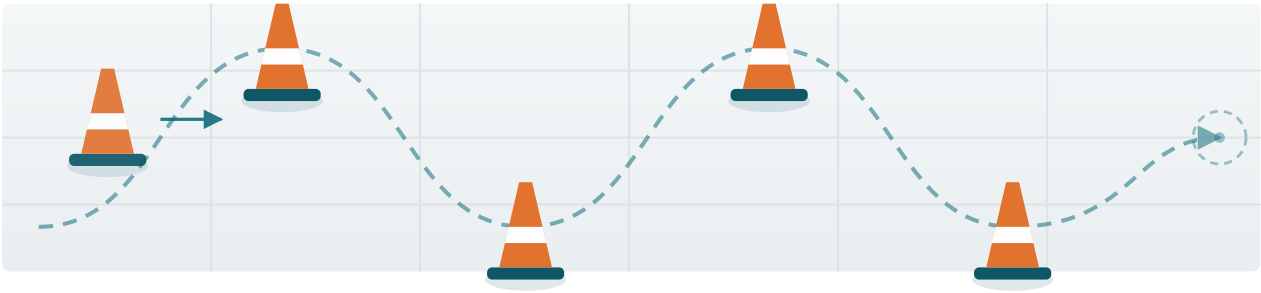


Collaborative Training Cone System

Functional Specification, System Architecture & Project Plan

A swarm of self-driving traffic cones for driver training. The instructor draws or selects an exercise on a tablet; the cones assign themselves to positions and drive there together without colliding. Selecting a different exercise rearranges the pad in under a minute.

53	15	7	6	2
FEATURES SPECIFIED	MEASURABLE REQUIREMENTS	STANDARD EXERCISES	CONES IN THE BASELINE FLEET	SEMESTERS TO DEFENCE



DOCUMENT	PROJECT ADVISOR	TEAM	PREPARED BY
Design Specification	Ahmad Koubeissi	1 × MECH + 3 × MECA	Mohamad Fayoumi
Revision 1.0 · 8 September 2026		Four vertical slices	

Contents

Executive Summary	1
Document Control	2
Part I — Functional Specification	3
1 Purpose and scope	3
2 Feature catalogue	4
3 Standard exercise presets and fleet sizing	7
4 Use cases	7
5 The tablet application	9
6 Non-functional requirements	9
7 Prioritisation and the minimum demonstrable product	10
8 Verification	10
9 Explicitly out of scope	11
10 Open questions for the advisor	11
Part II — System Architecture	12
11 The four decisions that define the architecture	12
12 Software decomposition	14
13 Recommended technology stack	15
14 Development strategy	15
Part III — Execution	16
15 Team split	16
16 Timeline	17
17 Sequencing rules	17
18 Risk register	18
19 What to do first	18
Appendices	19
A Feature index	19
B Derived parameters and design values	20
C Glossary	21

Reading order. Part I is the specification and is authoritative. Part II gives one defensible way of meeting it. Part III schedules the work in Part I's own terms. If you read only two sections, read **§7.2** — the four-minute demonstration the whole project is built toward — and **§10**, the five questions that must be put to the advisor before anything is built.

EXECUTIVE SUMMARY

FRONT MATTER

One page for the reader who will not read the other twenty-one: what is being built, what this document decides, what it does not yet decide, and what happens next.

§ The problem

A driving instructor sets each exercise by walking the training pad and placing cones by hand. That consumes ten to fifteen minutes of a sixty-minute lesson, produces geometry that is not repeatable between sessions, and puts an instructor on foot in the same space as a learner driver.

§ The system

A fleet of self-driving traffic cones. The instructor selects or draws an exercise on a tablet; the cones assign themselves to positions and drive there together without colliding. Selecting a different exercise rearranges the pad in under a minute. Cone geometry becomes repeatable to the centimetre and the instructor never leaves the vehicle.

§ What this document decides

QUESTION	DECISION	REASONING, IN BRIEF	SEE
Where does control live?	Hybrid — the coordinator assigns and plans globally; each cone controls its own pose and runs its own reflexes	Safety never crosses the radio, and the split maps cleanly onto four students	§11.1
How does a cone know where it is?	A UWB tag trilaterated against four surveyed anchors, fused with wheel odometry and a gyroscope in an EKF	The only method both outdoor-capable and affordable per unit. The project's highest-risk subsystem	§11.2
How do cones talk?	ESP-NOW for the control link through a dedicated ESP32 gateway; Wi-Fi only between tablet and coordinator	2–5 ms latency, no access point needed on the pad, and it sidesteps the shared-channel constraint rather than fighting it	§11.3
How do they avoid each other?	Central ORCA at 10 Hz with a 0.60 m radius, plus an independent braking reflex on each cone	Continuous and collision-free under its assumptions; the radius is <i>derived</i> from a 190 ms latency budget, not guessed	§11.4
How many cones?	Six as the prototype baseline; software with no fixed limit, verified in simulation at twelve	The seven standard exercises need between four and twelve. The proposal does not state a number, yet it sets cost and pad size	§3
What gets built first?	The simulator, before any hardware exists	Three of the four students can then progress without waiting on the machine shop	§14.1
Which stack?	Python coordinator, C++ on ESP32-S3, a browser-based tablet application	The fastest route for this team. ROS 2 was considered and argued against — that call reverses if a student already knows it	§13

§ What this document adds to the approved proposal

Four things, each argued where it appears: a **vehicle-present interlock** (F5.6), **re-deployment of a displaced cone** (F3.8), a **fixed fleet size**, and **measurable acceptance criteria** throughout. The first is a safety feature the proposal does not mention and that an assessment panel will ask about; the second is what makes the system usable in a setting where cones are routinely struck. See Document Control.

§ What is *not* yet decided

Five questions belong to the advisor, and all five should be closed before anything is built. They are set out in full in §10.

#	QUESTION	WHY IT CANNOT WAIT
1	Prototype fleet size	Sets the budget and the pad area
2	Pad dimensions, surface, and whether a specific site exists	Determines the localisation technology
3	Is an indoor demonstration acceptable?	The largest lever in the project. Indoors, an overhead camera gives ≈ 1 cm accuracy for the cost of one camera and the highest-risk subsystem becomes cheap and quick. Outdoors, that fails and UWB is required
4	Is heading control required, or is position enough?	Cones are rotationally symmetric. If heading does not matter, NFR-2 and a substantial slice of the localisation work disappear
5	Must a cone survive being driven over, or only bumped?	Drives the entire mechanical concept

THE SINGLE MOST USEFUL THING TO DO THIS WEEK

Put **questions 3 and 4** to the advisor, in that order, before any other work starts. Between them they can remove the majority of the project's technical risk at no cost to its academic value — and answered late, they invalidate work that has already been done.

§ What success looks like

THE TARGET — FOUR MINUTES, ON HARDWARE, AT THE DEFENCE

Four cones. Load the slalom preset, adjust the spacing, preview, execute — the cones cross the area simultaneously without colliding and settle. **READY** appears. Load a parking bay and execute again. Knock a cone over; the system detects it, alerts, and re-deploys it once righted. Press **E-STOP** mid-motion; everything halts.

That one sequence exercises effectively the whole *Must* feature set and tells the story of the problem without a slide. Build toward it specifically (§7.2).

§ Start here

1. **Close the five questions in §10 with the advisor** — questions 3 and 4 first.
2. **Fix the fleet size and the pad assumption**, and update §3 accordingly.
3. **Agree the two interfaces in §15 and write them down** — the cone message set, and the coordinator-to-cone abstraction. These are where four separate workstreams meet, and where integration projects usually fail.
4. **Start the simulator.** It is the longest pole with no dependencies.

Project	Collaborative Training Cone System
Document	Functional Specification, System Architecture & Project Plan
Revision	1.0 — first issue
Date	8 September 2026
Programme	Senior Learning Project (SLP), academic year 2026–2027
Category	Mechanical and Mechatronics
Advisor	Ahmad Koubeissi
Team composition	1 student (MECH) + 3 students (MECA)
Status	Design phase — specification and architecture. No implementation has begun, and none should begin before the five questions in §10 are closed.

§ How this document is organised

The document is in three parts and three appendices.

PART	TITLE	ANSWERS
I	Functional Specification	What the system does. Features, use cases, interface, measurable requirements, priorities and scope boundaries
II	System Architecture	How it does it. The design decisions, the reasoning behind each, and the resulting decomposition
III	Execution	Who builds what, and when. Team split, milestones, sequencing and risk
A–C	Appendices	Consolidated feature index, requirements index, glossary

PRECEDENCE

Part I is authoritative. It states what the system must do and is the basis on which the project will be assessed. Part II is subordinate: it is one defensible way of meeting Part I, and where the two conflict, the architecture changes rather than the specification. Part III schedules the work in Part I's own terms, so that progress is measured against the specification rather than against effort spent.

§ Identifier conventions

Every feature and requirement carries a stable identifier. These identifiers are the project's common vocabulary: they should be cited in commit messages, test case names, report sections and the defence presentation, so that any statement about progress points at something specific in this document.

PREFIX	MEANING	EXAMPLE
<i>F_n.m</i>	Functional feature	F3.6 — fleet READY indication
<i>NFR-<i>n</i></i>	Non-functional requirement	NFR-1 — position accuracy within 15 cm RMS
<i>UC-<i>n</i></i>	Use case	UC-3 — a cone is knocked over during an exercise
<i>RSK-<i>n</i></i>	Risk register entry	RSK-1 — outdoor UWB accuracy shortfall

§ Relationship to the approved project proposal

This document elaborates the approved SLP proposal. It preserves the proposal's objectives without exception and adds the detail needed to build and assess the system. Four substantive additions are made, each argued at the point it appears:

ADDITION	WHERE	WHY IT WAS ADDED
Vehicle-present interlock	F5.6 , §2.5	The system moves robots across a surface on which a learner driver is operating a car. An explicit, instructor-owned motion inhibit is the single most important safety feature and was not in the proposal
Displaced-cone re-deployment	F3.8 , §2.3	Training cones get struck. A system needing full redeployment after every contact would be worse than manual placement
Fleet size fixed at six	§3	The proposal leaves the number of cones open, yet it determines cost, pad size and the difficulty of coordination. The standard exercises require 4–12; six is the defensible prototype baseline
Measurable acceptance criteria	§6, §8	The proposal lists deliverables but no thresholds. Without numbers, “the cones reached their positions” cannot be assessed, defended, or failed

1 Purpose and scope

1.1 Problem statement

A driving instructor running a practical lesson on a training pad sets each exercise by walking the pad and placing traffic cones by hand, measuring spacings by pacing or with a tape. Changing from a slalom to a parallel-parking bay means walking the pad again.

Three costs follow. In a sixty-minute lesson, ten to fifteen minutes of instruction time is spent on cone placement rather than teaching. Geometry is inconsistent between sessions and between instructors, so a trainee's performance is not comparable from one lesson to the next. And placement puts an instructor on foot in the same space in which a learner driver is operating a vehicle.

1.2 What the system does

The Collaborative Training Cone System replaces manual placement with a fleet of self-driving cones. The instructor selects or draws an exercise on a tablet; the fleet arranges itself into that exercise; redrawing rearranges it. Cone geometry becomes repeatable to the centimetre, changeover takes under a minute, and the instructor never leaves the vehicle or the edge of the pad.

1.3 System boundary

INSIDE THE BOUNDARY	COMPRISING
Cone units	The mobile robots, their shells, drives, sensing, power and firmware
Localisation infrastructure	Fixed anchors and their survey, plus the development ground-truth rig
Radio gateway	The bridge between the coordinator host and the cone control link
Coordinator	The software that compiles layouts, assigns cones and plans motion
Tablet application	The instructor's entire interface to the system

Outside the boundary: the training pad and its surface, the training vehicle, the driving syllabus, mains power and charging infrastructure, and the tablet hardware itself.

1.4 Actors

ACTOR	TYPE	INTERACTION AND ASSUMPTIONS
Instructor	Primary human user	Authors and deploys layouts, monitors the fleet, stops it. Assumed non-technical, and assumed to be seated in or standing beside a vehicle
Technician	Secondary human user	First-time pad setup, calibration, charging, firmware updates, fault recovery. Assumed trained on the system
Cone unit	Autonomous agent	Localises, drives, avoids its neighbours, reports its state, and protects itself when contact with the outside world goes wrong
Coordinator	Software agent	Compiles layouts, assigns and plans, and mediates every instructor command
Trainee driver	Affected party — not a user	Never operates the system and may not be aware of it. Their safety constrains every requirement in §2.5

1.5 Assumed operating environment

An outdoor asphalt or concrete training pad, nominally 30 × 50 m, effectively flat (gradient ≤ 5%), in daylight, dry to lightly wet, at 0–45 °C, with no overhead cover and one training vehicle present at a time.

CONFIRM BEFORE DESIGN FREEZE

These assumptions drive a large fraction of the requirements that follow — in particular the choice of localisation technology, which is the project's highest-risk subsystem. They are stated here as assumptions precisely so that they can be checked. See §10, question 2.

2 Feature catalogue

Fifty-three features in eight functional areas. Each carries an identifier that is used throughout the remainder of the project.

F1 Course authoring - Freehand path & primitives - Spacing in metres, per segment - Numeric dimensioning - Live feasibility check	F2 Layout library - Save, load, manage - Seven standard exercises - Lesson sequences - Import / export	F3 Deployment - Preview with ETA - Execute, pause, abort - Unambiguous READY - Gather & re-deploy	F4 Monitoring - Live map, deviation - Battery & link health - Alerts - Session event log
F5 Safety MANDATORY - E-stop, 500 ms watchdog - Geofence - Tip & obstacle reflex - Vehicle-present interlock	F6 Cone autonomy - Self-localise - Drive to pose, hold it - Mutual avoidance - Self-test, battery, identify	F7 Administration - Pad survey & boundary - Fleet registration - Guided calibration - Diagnostics & updates	F8 Data - Session record - Accuracy record - CSV / JSON export - Evidence for the report

Figure 1. The eight functional areas. F5 is shown apart because every feature in it is mandatory and none may be traded away for schedule.

2.1 F1 — Course authoring

The instructor defines exercise geometry on the tablet.

ID	FEATURE	DESCRIPTION
F1.1	Freehand path	Draw a continuous path with a finger or stylus on a scaled plan of the pad
F1.2	Point placement	Tap to place, drag to move, long-press to delete an individual cone position
F1.3	Geometric primitives	Insert a parameterised straight line, arc, circle or rectangle instead of drawing by hand
F1.4	Path editing	Insert, move and delete control points; unlimited undo and redo
F1.5	Automatic spacing	Distribute cones evenly along a path at a spacing given in metres — default 2.0 m, range 0.5–20 m, settable per segment
F1.6	Numeric dimensioning	Enter exact dimensions instead of drawing: a 6.0 × 2.5 m parking bay, or a slalom at 9.0 m pitch
F1.7	Feasibility check	Validate continuously <i>while the layout is being drawn</i> : cones required against cones available and healthy, every point inside the pad boundary, no two points closer than the minimum separation. Violations are reported in place
F1.8	Layout transform	Translate, rotate and scale the whole layout on the pad; mirror about an axis
F1.9	Reference overlay	Show pad boundary, a metric grid, and optionally the previous layout in grey for comparison

DESIGN RATIONALE — F1.7

This is the feature that prevents the most common operational failure. Without it, an instructor discovers *mid-deployment* that the layout needs nine cones and the fleet has six healthy ones — at which point cones are already moving and the exercise is late. Validation belongs at authoring time, not at execution time.

2.2 F2 — Layout library

ID	FEATURE	DESCRIPTION
F2.1	Save	Store the current layout under a name, with author and date
F2.2	Load	Recall a stored layout into the editor, or deploy it directly
F2.3	Standard presets	Ship a built-in, parameterised set (S3) so a new instructor is productive without drawing anything
F2.4	Manage	Duplicate, rename, delete; mark favourites
F2.5	Import / export	Exchange layouts between instructors and installations as a documented JSON file
F2.6	Lesson sequence	Group layouts into an ordered lesson; a single Next Exercise action advances the fleet from one to the next

DESIGN RATIONALE — F2.6

This feature expresses the value of the whole project better than any other. A single button that reconfigures an entire training pad between exercises makes the case to an assessment panel far more effectively than any one layout can. It is inexpensive to build once F2.2 and F3.2 exist.

2.3 F3 — Deployment and execution

ID	FEATURE	DESCRIPTION
F3.1	Preview	Before any cone moves, show which cone is assigned to which target and the path each will take, with an estimated time to completion

ID	FEATURE	DESCRIPTION
F3.2	Execute	Commit the layout; the fleet moves
F3.3	Live progress	Per-cone state and remaining distance, an overall percentage, and an updating ETA
F3.4	Pause / resume	Freeze the fleet in place and continue from the same assignment
F3.5	Abort	Stop the operation, offering stop-in-place or return-to-previous-layout
F3.6	READY signal	An unambiguous fleet-wide indication — on the tablet and on the cones — that every cone is within tolerance and has settled, so the instructor knows the exercise may begin
F3.7	Gather	Send every cone to a defined collection area for storage or charging
F3.8	Re-deploy displaced	Detect a cone knocked out of position and, on instructor confirmation, drive it back to its target without disturbing the rest of the layout

SAFETY RATIONALE — F3.6

READY is a safety feature, not a convenience. The instructor must never have to guess whether the fleet has finished moving before releasing a vehicle onto the pad. It is specified as a full-width banner rather than a small icon for exactly this reason (§5).

DESIGN RATIONALE — F3.8

Cones in driver training get hit; that is what they are for. A system that requires a full redeployment after every contact is worse than manual placement, because the instructor must now interrupt the lesson and wait for six cones. A system that quietly nudges one cone back into place is better than anything a human can do without stopping the vehicle.

2.4 F4 — Monitoring

ID	FEATURE	DESCRIPTION
F4.1	Live map	Real-time cone positions on the pad plan, with target positions shown as ghosts and deviation drawn as a vector
F4.2	Per-cone status	Identifier, state, battery percentage, link quality, position error, and time since last contact
F4.3	Fleet summary	Count of cones ready, moving, faulted and offline, readable at a glance
F4.4	Alerts	Raise and clear: low battery, link lost, cone tipped, cone blocked or stalled, localisation degraded, geofence violation
F4.5	Event log	Timestamped record of every command, state change and alert in the session

2.5 F5 — Safety

Every requirement in this group is mandatory. None may be traded away for schedule, and none may be deferred to a later revision.

ID	FEATURE	DESCRIPTION
F5.1	Emergency stop	A single action stops the entire fleet. Available as a permanent on-screen control <i>and</i> as a physical handheld button independent of the tablet
F5.2	Comms watchdog	A cone that receives no valid command for 500 ms stops itself. Safety does not depend on the radio working
F5.3	Geofence	A cone refuses to drive outside the configured pad boundary regardless of the commanded target
F5.4	Tip-over detection	A cone that is knocked over cuts motor drive within 1 s and reports it
F5.5	Obstacle reflex	A cone that detects an object within 40 cm ahead brakes, independently of the central planner
F5.6	Vehicle-present interlock	While the instructor has flagged a vehicle as active on the pad, the fleet will not move. Overriding requires a deliberate, separate confirmation
F5.7	Speed limiting	Configurable maximum speed, enforced in firmware, defaulting to 0.8 m/s
F5.8	Movement conspicuity	A moving cone shows a flashing light and sounds an intermittent tone

ADDITION TO THE PROPOSAL — F5.6

The vehicle-present interlock does not appear in the approved proposal and should be added to it. The system moves robots across a surface on which a learner driver is operating a car. An explicit interlock, owned by the instructor rather than inferred by the system, is the single most important safety feature in this specification — and it is the first thing an assessment panel will ask about.

Note the deliberate division of responsibility: F5.1, F5.6 and F5.7 are policy, set by a human. F5.2, F5.3, F5.4 and F5.5 are reflexes, implemented on the cone. The reflexes must remain effective when the coordinator, the radio or the tablet has failed. No safety property in this specification depends on a working radio link.

2.6 F6 — Cone autonomy

Functions each cone performs on its own.

ID	FEATURE	DESCRIPTION
F6.1	Self-localisation	Continuously estimate its own position and heading on the pad
F6.2	Drive to pose	Navigate to a commanded position, and to a commanded heading where the exercise requires one
F6.3	Hold position	Maintain position against wind, slope and minor contact; correct drift automatically
F6.4	Mutual avoidance	Avoid other cones while in motion, cooperatively rather than by stopping
F6.5	Telemetry	Report pose, state, battery and health at 10 Hz
F6.6	Power-on self-test	Verify motors, encoders, IMU, radio and localisation at boot; refuse to join the fleet on failure

ID	FEATURE	DESCRIPTION
F6.7	Battery management	Report state of charge; warn at 20%; refuse new assignments below 10%; return to the gather point while still able
F6.8	Identify	On request, blink and beep so an operator can pick one physical cone out of the fleet

DESIGN RATIONALE — F6.8

Identify costs almost nothing and will be used constantly. The moment there are six identical cones on a pad and cone 4 reports a fault, somebody has to work out which one cone 4 is. Without this feature that means reading serial numbers off the underside of six units.

2.7 F7 — Administration

ID	FEATURE	DESCRIPTION
F7.1	Pad definition	Define the coordinate origin, the boundary polygon, the gather area, and the surveyed anchor positions
F7.2	Fleet registration	Add, remove and rename cone units; assign identifiers
F7.3	Calibration	Guided routines for wheel odometry scale, wheel track width, IMU bias, and anchor position refinement
F7.4	Firmware update	Update every cone in the fleet from the coordinator, with version reporting
F7.5	Diagnostics	Per-cone raw sensor view and manual jog control, for bring-up and fault finding
F7.6	Defaults	Set default spacing, speed limits, position tolerance and geofence margin

2.8 F8 — Data

ID	FEATURE	DESCRIPTION
F8.1	Session record	Which layouts were deployed, when, and how long each changeover took
F8.2	Accuracy record	Commanded versus achieved position for every deployment
F8.3	Export	Emit session and accuracy data as CSV or JSON

DESIGN RATIONALE — F8.2

The accuracy record exists for the capstone report as much as for the product. It is the mechanism that produces measured evidence for NFR-1 and NFR-3 in the verification matrix (§8). Build it early and *every test run from that point on generates report material automatically*, with no extra effort and no manual measurement.

3 Standard exercise presets and fleet sizing

These presets define the geometry the system must be able to produce, and therefore fix the minimum fleet size and the required pad dimensions. All parameters are adjustable; the values given are the defaults shipped with the system.

PRESET	GEOMETRY	CONES	DEFAULT PARAMETERS
Slalom	Single line of equally spaced cones	5	Pitch 9.0 m (range 5–15 m), entry offset
Parallel parking	Bay marked by four corner cones, optional two lead-in	4–6	Bay 6.0 × 2.5 m
Reverse bay parking	Two bay-defining lines	6	Bay 5.0 × 2.4 m, approach lane 4.0 m
Three-point turn	Two parallel kerb lines	6	Road width 6.0 m, length 12 m
Emergency lane change	Offset gates, ISO 3888-2 style	8–12	Gate widths derived from vehicle width; sections 12 / 13.5 / 11 / 12.5 / 12 m
Circle / figure-of-eight	One or two circles	8–12	Radius 8.0 m
Garage / 90° park	Three-sided bay	5	5.0 × 2.5 m

CONSEQUENCE — FLEET SIZE

The approved proposal does not state how many cones the fleet contains, yet that number determines cost, the pad area required, and the difficulty of the coordination problem. The presets above require between four and twelve.

A fleet of six is adopted as the prototype baseline. It covers the majority of the presets outright and degrades gracefully on the rest by increasing spacing. Two constraints follow, and both are binding on Part II: the software must impose *no fixed fleet limit* and must be verified in simulation at twelve (NFR-14); and where a layout needs more cones than are available, F1.7 must say so at authoring time and offer to increase the spacing instead of failing.

4 Use cases

UC-1 — Set up an exercise before a lesson

Actor: Instructor · **Precondition:** Fleet powered, localised and in the gather area · **Frequency:** Several times per day

1. The instructor opens the tablet application; the live map shows six cones ready.
2. The instructor opens the layout library and selects **Slalom**. **F2.3**
3. The instructor adjusts the pitch from 9.0 m to 12.0 m; the preview updates live. **F1.5**, **F1.6**
4. The system validates: five cones needed, six available, all points inside the boundary. **F1.7**
5. The instructor taps **Preview**; the system shows the assignment and an estimated 38 s. **F3.1**
6. The instructor confirms no vehicle is on the pad and taps **Execute**. **F3.2**, **F5.6**
7. Cones move; the map shows live progress. **F3.3**
8. All cones settle within tolerance; **READY** is displayed and the cones stop flashing. **F3.6**
9. The instructor begins the lesson.

ALTERNATE FLOWS

- 4a. Fewer healthy cones than the layout needs → the system offers to increase the spacing or deploy a reduced version, and will not offer Execute until this is resolved.
- 6a. The vehicle interlock is set → Execute is disabled, with the reason shown.
- 7a. A cone stalls → an alert is raised; the remaining cones complete; the instructor may retry that cone, exclude it, or abort.

UC-2 — Change exercise mid-lesson

Actor: Instructor · **Precondition:** A layout is deployed and a lesson is in progress · **This is the project's headline demonstration**

1. The instructor stops the vehicle clear of the pad and sets the vehicle interlock to *clear*. **F5.6**
2. The instructor taps **Next Exercise** in the active lesson sequence. **F2.6**
3. The system computes the assignment from current positions to the new layout; cones already near a target keep it.
4. The instructor taps **Execute**; the fleet reconfigures in under 60 s. **NFR-3**
5. **READY** appears and the lesson continues.

WHY THIS ONE MATTERS

Every other use case demonstrates a capability. This one demonstrates the *benefit* — the ten to fifteen minutes of lesson time recovered in §1.1. It should be the sequence performed live at the defence.

UC-3 — A cone is knocked over during an exercise

Actors: Cone, then Instructor · **Trigger:** Vehicle contact · **Frequency:** Routine, not exceptional

1. A vehicle contacts a cone; the cone tips.
2. The cone detects the tilt within 1 s, cuts motor drive and reports **TIPPED**. **F5.4**
3. The tablet raises an alert identifying which cone, and shows it on the map. **F4.4**
4. The instructor pauses the lesson and rights the cone by hand.
5. The cone re-localises and reports **IDLE**.
6. The instructor taps **Re-deploy**; only that cone returns to its target. **F3.8**
7. **READY** returns.

DESIGN CONSEQUENCE

This use case imposes a hard constraint on Part II: **righting a cone by hand must be sufficient to recover it**. No tablet interaction, no power cycle, no re-pairing, no technician. Any design in which recovery needs more than step 4 fails this use case.

UC-4 — A cone's battery runs out mid-session

Actors: Cone, Coordinator, Instructor · **Precondition:** A session is in progress

1. The cone reaches 20% charge → a warning alert is raised; the cone continues. **F6.7**
2. The cone reaches 10% → it refuses further assignments, informs the coordinator and requests recall.
3. The coordinator excludes it from future assignments and warns the instructor that the effective fleet is now five.
4. The instructor sends that cone to gather; F1.7 now validates all subsequent layouts against five cones.

DESIGN CONSEQUENCE

Fleet size is a *runtime* quantity, not a configuration constant. Every feature that depends on it — F1.7 above all — must read the count of currently healthy cones rather than the count of registered ones.

UC-5 — End of day

Actors: Instructor, Technician

1. The instructor taps **Gather**. **F3.7**
2. All cones drive to the collection area, arrange themselves in a row and stop.
3. The technician switches them off and puts them on charge.

UC-6 — First-time installation on a new pad

Actor: Technician · **Frequency:** Once per site, plus after any anchor is moved

1. Mount four anchors on poles at the pad corners and survey their positions.
2. Enter the anchor coordinates, define the origin, trace the boundary and mark the gather area. **F7.1**
3. Register each cone unit. **F7.2**
4. Run the guided calibration routines. **F7.3**
5. Run the verification layout and confirm the achieved accuracy meets §6. **F8.2**

5 The tablet application

Five screens. From anywhere in the application, the instructor must be able to reach an emergency stop without navigating — F5.1 is present on every screen, not on one of them.

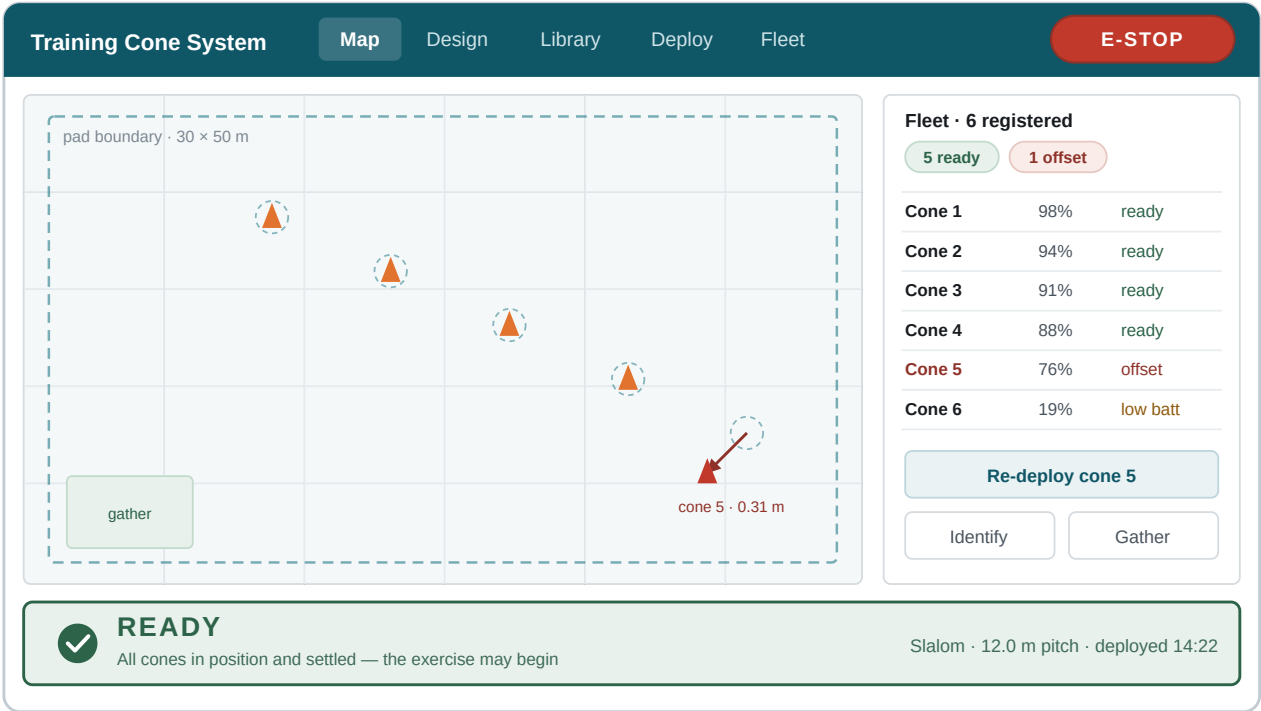


Figure 2. The Map screen, the application's home. E-STOP is permanently present in the top bar (F5.1); READY is a full-width banner rather than an icon (F3.6); cone 5 has been displaced and its deviation vector and one-tap re-deploy are both visible (F3.8, F4.1). Cone 6 is below the 20% warning threshold (F6.7), so F1.7 will validate the next layout against five cones, not six.

SCREEN	CONTAINS	FEATURES
Map home	Scaled plan of the pad, live cone positions, fleet status strip, persistent E-STOP	F4.1, F4.2, F4.3, F5.1
Design	Drawing canvas, tool palette, spacing and dimension inputs, live feasibility banner	F1.1 – F1.9
Library	Presets, saved layouts, lesson sequences	F2.1 – F2.6
Deploy	Assignment preview, ETA, Execute / Pause / Abort, vehicle interlock toggle, READY banner	F3.1 – F3.8, F5.6
Fleet	Per-cone detail, identify, alerts, event log, entry to diagnostics	F4.4, F4.5, F6.8, F7.5

5.1 Interaction constraints

These follow from the operating environment in §1.5, not from taste. Each one is testable and each one is a defect if violated.

- **Usable in direct sunlight.** High contrast throughout; large type; no thin light-grey text on white.
- **Usable with gloves and wet hands.** Touch targets no smaller than 48 px; no hover states; no gestures requiring precision or two hands.
- **Usable one-handed** while standing beside a vehicle.
- **No destructive action without confirmation** — with exactly one exception. E-STOP must be instant and must never ask.
- **Fleet state readable in under two seconds** from across a vehicle cabin. This is the reason READY is a banner and not an icon, and the reason the fleet summary is a row of counts rather than a list to be read.

6 Non-functional requirements

These are the thresholds against which the system will be measured. §8 states how each is proved.

ID	CATEGORY	REQUIREMENT
NFR-1	Accuracy	Achieved position within 15 cm RMS of commanded, and within 25 cm at the 95th percentile
NFR-2	Accuracy	Achieved heading within 10° , where a heading is specified
NFR-3	Performance	Full reconfiguration of six cones over a 20 × 20 m area in ≤ 60 s
NFR-4	Performance	Instructor command to first cone movement ≤ 500 ms
NFR-5	Performance	Live map updates at ≥ 5 Hz
NFR-6	Reliability	Command link packet loss < 2% at 50 m line of sight
NFR-7	Reliability	Zero cone-to-cone collisions in 20 consecutive reconfigurations

ID	CATEGORY	REQUIREMENT
NFR-8	Endurance	≥ 2 h of realistic session duty per charge
NFR-9	Safety	Fleet halts within 500 ms of E-STOP or of loss of communications
NFR-10	Environmental	Operates 0–45 °C, in daylight, on dry to lightly wet asphalt
NFR-11	Environmental	Survives repeated low-speed vehicle contact and being driven over; no exposed rigid edges
NFR-12	Environmental	Ingress protection to at least IP54
NFR-13	Usability	An instructor with no technical background deploys a preset within 5 minutes of first use, unaided
NFR-14	Scalability	Software imposes no fixed fleet limit; verified in simulation at twelve cones
NFR-15	Maintainability	A cone can be opened and its battery replaced with hand tools in under 5 minutes

EMPHASIS — NFR-11

The proposal asks for "a stable and durable mechanical structure that protects the internal components while maintaining the appearance and function of a training cone." In engineering terms that means three things: the shell is **sacrificial and compliant**, the electronics are **mechanically isolated** from shell deformation, and the centre of mass is low enough that the unit either stays upright or is trivially righted by hand — a requirement that UC-3 makes binding. The severity of this requirement is the single largest driver of the mechanical design, and it is open question 5 in §10.

7 Prioritisation and the minimum demonstrable product

7.1 MoSCoW classification

PRIORITY	FEATURES
MUST no project without these	F1.1, F1.5, F1.7, F2.1, F2.2, F2.3, F3.1, F3.2, F3.3, F3.6, F4.1, F4.2, F5.1, F5.2, F5.3, F5.4, F5.7, F6.1, F6.2, F6.3, F6.4, F6.5, F6.6,
SHOULD expected in a complete system	F1.2, F1.3, F1.4, F1.6, F1.8, F2.4, F3.4, F3.5, F3.7, F3.8, F4.3, F4.4, F4.5, F5.5, F5.6, F5.8, F6.7, F6.8, F7.5, F8.1, F8.2
COULD if time permits	F1.9, F2.5, F2.6, F7.4, F7.6, F8.3
WON'T not this project cycle	See §9

TWO DELIBERATE ANOMALIES IN THIS TABLE

F2.6 is classified *Could* on effort, but it is the strongest demonstration feature in the catalogue and is inexpensive once F2.2 and F3.2 exist. Promote it the moment the Must set closes — it is what UC-2 needs.

F5.6 is classified *Should* only because it is trivial to implement, not because it is optional. Every feature in F5 is mandatory (§2.5). The MoSCoW column measures build effort and sequencing; it does not grant permission to omit a safety feature.

7.2 Minimum demonstrable product

The subset that must work live, on hardware, at the defence:

THE FOUR-MINUTE SEQUENCE

Four cones. The instructor loads the slalom preset, adjusts the spacing, previews and executes. The cones cross the area simultaneously without colliding and settle within tolerance. **READY** appears. The instructor loads a parking bay and executes again; the cones reconfigure. A cone is deliberately knocked over; the system detects it, alerts, and re-deploys it once it has been righted by hand. The instructor presses **E-STOP** mid-motion and everything halts.

That single sequence exercises F1.5, F1.7, F2.3, F3.1, F3.2, F3.6, F3.8, F4.1, F4.4, F5.1, F5.4 and F6.1 – F6.4 — effectively the whole Must set — in about four minutes, and it tells the story of §1.1 without a slide. **Build toward this sequence specifically, and rehearse it.** Anything that does not serve it, or serve the report, is not on the critical path.

8 Verification

Every requirement needs a stated method of proof. "It looked like it worked" is not one, and neither is a single successful run. The evidence column names the artefact that ends up in the report.

REQUIREMENT	METHOD	EVIDENCE PRODUCED
NFR-1, NFR-2	30 deployments measured against the overhead-camera ground-truth rig (§14.2)	Error scatter plot and cumulative distribution, generated from F8.2 logs
NFR-3	10 timed reconfigurations; worst case recorded, not the mean	Timing table
NFR-4, NFR-5	Instrumented timestamps end to end	Latency histogram
NFR-6	Packet counters at 10, 25 and 50 m	Loss-versus-distance chart
NFR-7	20 consecutive reconfigurations, video reviewed frame by frame	Video plus event log

REQUIREMENT	METHOD	EVIDENCE PRODUCED
NFR-8	Duty-cycle discharge test to the 10% cut-off	Discharge curve
NFR-9	20 E-STOP events and 20 induced communications failures, timed	Stop-latency table
NFR-13	Two instructors unfamiliar with the system, timed, unaided, not coached	Observation notes and completion times
NFR-14	Simulated fleet of twelve	Simulation run log
F5.3, F5.4, F5.5	Deliberate provocation of each condition, 10 trials each	Pass / fail table

SEQUENCING CONSEQUENCE

Two pieces of infrastructure in this table — the ground-truth rig and the F8.2 accuracy log — are prerequisites for most of the evidence, not by-products of it. §17 therefore requires both to exist by week 8 of the first semester, long before the measurements themselves are needed.

9 Explicitly out of scope

Stating these protects the project from scope creep and pre-empt's the panel's questions. Each is a reasonable extension; none is in this cycle.

EXCLUDED	REASONING
Autonomous obstacle mapping	The pad is assumed open and known. Cones avoid <i>each other</i> , not arbitrary unmapped obstacles
Vehicle detection or tracking	The system does not sense the training vehicle. The interlock (F5.6) is set by the instructor, never inferred — an inferred interlock that fails is worse than none
Operating while a vehicle moves on the pad	Cones move only between exercises. This is a safety boundary, not a capability gap
Automatic charging or docking	Gather (F3.7) brings cones to a collection point; a human connects the charger
Night, heavy rain, snow or ice	Outside the assumed environment of §1.5
Multiple simultaneous instructors or pads	One coordinator serves one pad and one instructor
Kerbs, grass or gravel	The drivetrain is specified for flat paved surfaces only
Regulatory certification	The unit is not certified as road furniture

10 Open questions for the advisor

These five questions change the specification. **They should be closed before the design freeze in week 2**, because each of them invalidates work if answered late.

#	QUESTION	WHAT CHANGES WITH THE ANSWER
1	What is the prototype fleet size? §3 argues for six.	Budget, pad area, and the difficulty of the coordination problem. If the budget covers fewer, the balance is simulated and §7.2 still holds — it needs only four
2	What are the pad dimensions and surface, and is there a specific site?	The choice of localisation technology, which is the project's highest-risk subsystem (§11.2)
3	Is an indoor demonstration acceptable?	The largest single lever in the project. Indoors, an overhead camera gives roughly 1 cm accuracy for the cost of one camera, and the highest-risk subsystem becomes cheap, accurate and quick. Outdoors, that fails and UWB is required
4	Is heading control required, or is position sufficient?	Cones are rotationally symmetric, so heading matters only for asymmetric markers. If it is not required, NFR-2 and a substantial slice of the localisation work disappear
5	Must a cone survive being driven over, or only bumped?	The severity of NFR-11, and with it the entire mechanical concept — shell material, isolation strategy and mass budget

RECOMMENDED ACTION

Questions 3 and 4 together can remove the majority of the project's technical risk at no cost to its academic value. They should be put to the advisor first, and in that order.

Part I states what the system must do. This part states one defensible way of achieving it, and argues each choice against the alternatives that were rejected. The requirements are not restated here; §6 and §8 remain the reference.

11 The four decisions that define the architecture

Everything else follows from these four. Each is argued rather than asserted, because a defence panel will ask why not the other one.

11.1 Where does the intelligence live?

OPTION	ASSESSMENT
Fully centralised coordinator computes wheel speeds; cones are radio-controlled motors	Simple to reason about, but every packet loss becomes a control-loop dropout, and F5.2 cannot be satisfied on the cone because the cone has no notion of its own goal
Fully decentralised each cone runs local swarm rules — potential fields, flocking	Superficially the more impressive “swarm” answer, but position assignment becomes emergent and therefore unrepeatable. A driving exercise is a <i>specified</i> geometry, not an emergent one, and NFR-1 is stated in centimetres
Hybrid coordinator does assignment and global avoidance; cone does pose control and reflexes	Chosen. See below

The coordinator owns everything global — who goes where, whose path yields to whose. Each cone owns everything local and fast: wheel PID, goal seeking, emergency stop, tip detection, communications watchdog.

WHY THIS SPLIT, SPECIFICALLY

Safety does not cross the radio. If the coordinator dies, cones coast to a stop on their own watchdog (F5.2). Every reflex in §2.5 remains effective with the link down, which is exactly the property NFR-9 demands.

It maps onto the team. One student can own the coordinator without touching firmware, and another can own firmware without touching the planner — see §15. An architecture that forces two students into the same file every week is a scheduling hazard, not just an engineering one.

11.2 How does a cone know where it is?

This is the highest-risk subsystem in the project. If localisation is poor, nothing downstream can recover — NFR-1, NFR-3 and NFR-7 all rest on it — so it warrants the most analysis and the earliest measurement.

METHOD	ACCURACY	COST / CONE	OUTDOORS	HEADING	WORKING AREA
Wheel odometry + IMU alone	drifts without bound	≈ \$10	yes	yes, drifting	unlimited
Overhead camera + ArUco markers	1–3 cm	≈ \$0 one camera total	poorly sun, shadow	yes, directly	≈ 12 × 9 m per camera
UWB trilateration (DW3000)	10–30 cm	≈ \$20	yes	no	≈ 100 m
RTK-GPS (u-blox ZED-F9P)	2 cm	≈ \$250	yes needs sky view	no	unlimited
Ultrasonic / IR beacons	10–50 cm, noisy	≈ \$15	marginal	no	≈ 10 m

The environment assumed in §1.5 — 30 × 50 m of open asphalt in daylight — eliminates the overhead camera as the *deployed* solution and makes RTK-GPS unaffordable across six or more units. UWB is the only remaining option that is both outdoor-capable and cheap enough to fit to every cone.

Chosen: a DW3000-class UWB tag on each cone, trilaterated against four surveyed anchors, fused with wheel odometry and a gyroscope in an Extended Kalman Filter.

Three consequences the team must plan around

CONSEQUENCE	DETAIL AND MITIGATION
UWB gives position, not heading	Recover heading from the gyroscope, corrected by the direction of travel — a differential-drive robot moving forward is, by construction, pointing where it is going. Correct gyro bias whenever speed exceeds 0.3 m/s. <i>Contingency:</i> a second tag 25 cm from the first gives heading directly; it doubles tag cost, so it is the fallback, not the baseline. Open question 4 in §10 may remove this problem entirely
Update rate falls as cones are added	With two-way ranging, one fix costs roughly 3 ms per anchor, so about 12 ms for four anchors. Under TDMA that is ≈ 13 Hz at six cones and ≈ 8 Hz at ten. Acceptable for the target fleet. <i>Scaling path:</i> migrate to TDoA — tags transmit, synchronised anchors listen — which scales to 100+ tags. Document this in the report; forward-looking analysis of a limit you have measured is worth more than avoiding the limit
Cones are low	A cone stands about 50 cm tall, so its tag sits near the ground where multipath off asphalt is worst. Mount anchors on 3–4 m poles, and never place all four near-collinear — poor geometry (high GDOP) ruins trilateration even with perfect range measurements

USE THE CAMERA ANYWAY — AS AN INSTRUMENT, NOT AS THE PRODUCT

During development, an indoor rig with a 1080p camera at 6 m and an ArUco marker on each cone lid yields roughly 1 cm absolute pose. That is the reference against which the UWB + EKF stack is *measured* to produce the numbers NFR-1 requires. A 90° camera at 6 m covers about 12 × 9 m; at 1920 px across 12 m, one pixel is 6.25 mm, and sub-pixel marker fitting does the rest.

Building a validation instrument, rather than eyeballing a tape measure, is what separates a strong capstone from an average one — and it is the same rig that answers open question 3 if the project moves indoors.

11.3 How do cones talk?

OPTION	LATENCY	RANGE	PAYLOAD	NOTE
ESP-NOW	2–5 ms	200 m LOS	250 B	Connectionless, needs no access point, native broadcast
Wi-Fi UDP via an AP	10–50 ms, jittery	≈ 50 m	large	Requires infrastructure in the middle of an asphalt pad
BLE	20–100 ms	30 m	small	Connection-oriented; poor for one-to-many
LoRa	100–1000 ms	kilometres	tiny	Far too slow for a 10 Hz control loop

ESP-NOW wins the control link outright. But the tablet needs ordinary Wi-Fi, and an ESP32 cannot sit on an access point's chosen Wi-Fi channel while running ESP-NOW on a different one: **ESP-NOW and Wi-Fi station mode must share a radio channel**. Fighting that constraint in firmware is a classic and avoidable time sink.

Chosen: do not fight it — separate the radios physically.

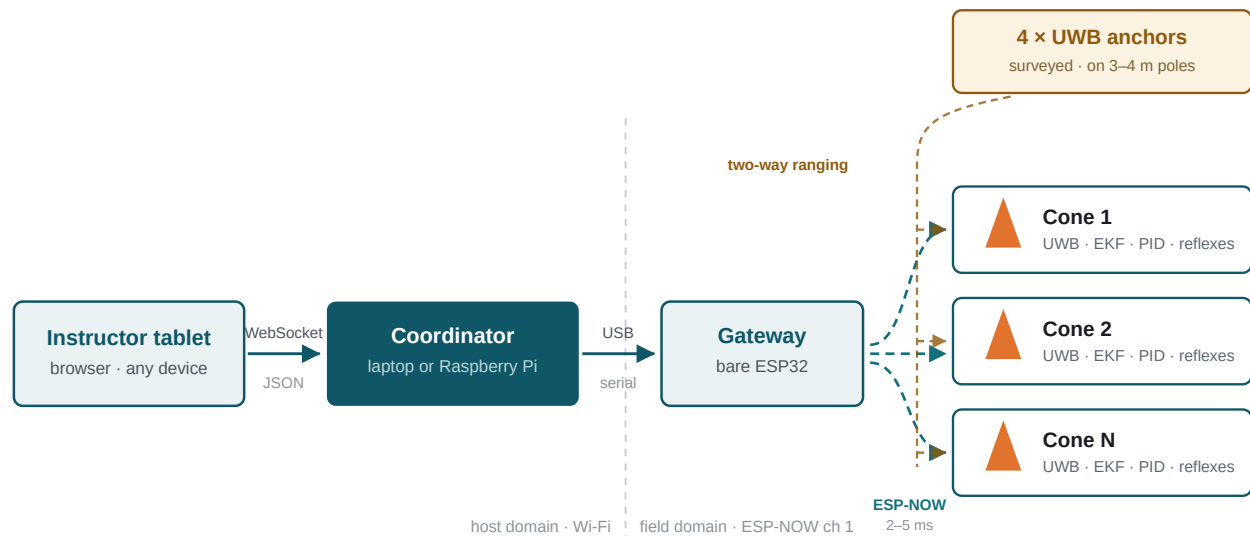


Figure 3. System topology. The gateway is a bare ESP32 doing one job — serial frames in, ESP-NOW out, and back — so it owns the radio channel exclusively while the coordinator host is free to serve Wi-Fi to the tablet. Each half is independently testable, which matters when two different students own them (§15).

11.4 How do cones avoid each other?

Six cones crossing a 20 m pad simultaneously is a genuine multi-robot motion planning problem, not an implementation detail. NFR-7 permits zero collisions.

APPROACH	ASSESSMENT
Artificial potential fields	Quick to write, but suffers local minima and oscillates in narrow passes. Not defensible against a panel that knows the literature
Prioritised / sequenced planning one cone moves at a time	Trivially safe, and trivially fails NFR-3: six cones crossing 20 m one after another exhausts the 60 s budget on travel alone
Time–space reservation grid plus time windows	Sound, but the discretisation is coarse relative to a 15 cm accuracy target and replanning after a disturbance is expensive
ORCA — reciprocal velocity obstacles	Chosen. Continuous rather than discretised, runs at kilohertz rates for six agents, collision-free under its stated assumptions, and is the standard citation in the field

ORCA assumes holonomic agents. Differential-drive cones are not holonomic, so the ORCA output velocity is wrapped by a unicycle controller that converts a desired (v_x, v_y) into (v, ω) with a bounded turn rate — the standard treatment, in which the effective agent radius is inflated to cover the turning transient.

Size the safety radius from the measured latency, not from a guess

UWB fix interval		≈ 80 ms
Coordinator cycle		100 ms
ESP-NOW + firmware		≈ 10 ms
Worst-case staleness		190 ms

At 0.8 m/s, 190 ms is 15 cm of travel the planner cannot see. The safety radius must cover it:

DERIVED PARAMETERS

$r_{\text{safety}} = r_{\text{cone}} (0.20 \text{ m}) + v \cdot t_{\text{stale}} (0.15 \text{ s}) + \text{margin} (0.20 \text{ m}) \approx 0.55 \text{ m} \rightarrow \text{use } \mathbf{0.60 \text{ m}}$

$v_{\text{max}} = \mathbf{0.80 \text{ m/s}}$ during coordinated motion, reduced to $\mathbf{0.30 \text{ m/s}}$ within 1 m of the goal (which also serves NFR-1, since terminal accuracy degrades with approach speed)

Independently of all of this, each cone runs a reflex layer: if the forward time-of-flight sensor reads under 40 cm, brake — regardless of what the coordinator says (F5.5). **ORCA is the plan; the reflex is the seatbelt.**

12 Software decomposition

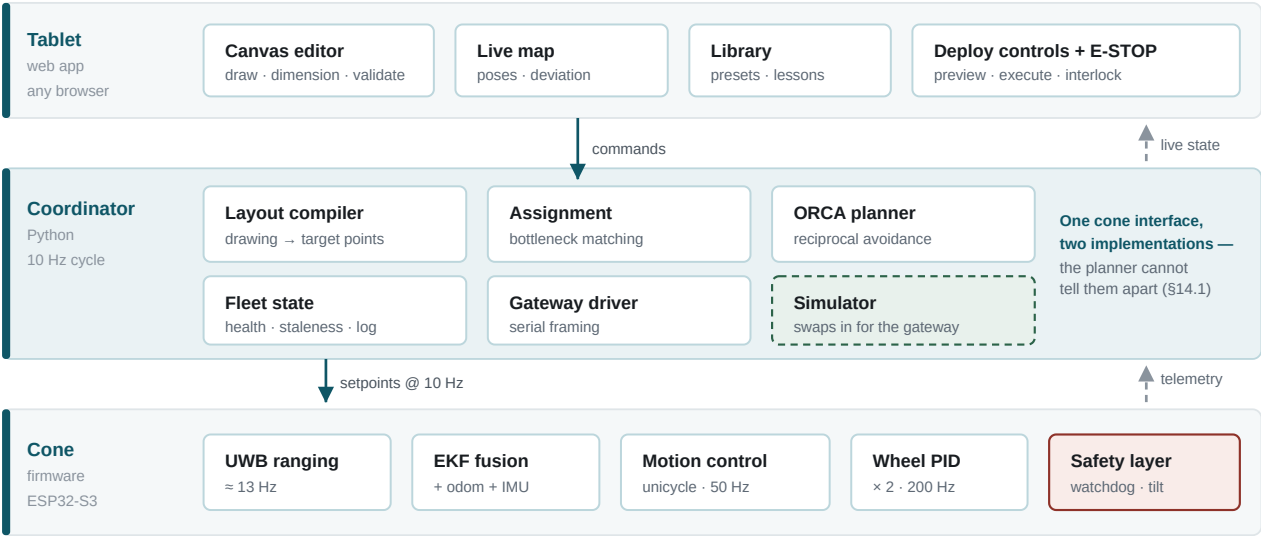


Figure 4. Software decomposition across the three tiers. The green dashed block is the point of the whole arrangement: the simulator and the real gateway implement the same cone interface, so the coordinator, the planner and the entire tablet application can be developed and tested with no hardware present (§14.1).

12.1 Coordinator responsibilities

MODULE	INPUT	OUTPUT	METHOD
Layout compiler	Polyline or spline from the tablet, in pad coordinates	Ordered list of target points	Arc-length resampling at the configured spacing (default 2.0 m); rejects the layout when points required exceed cones available (F1.7)
Assignment	N cone poses, $M \leq N$ targets	Cone → target mapping	Bottleneck (min–max) assignment — see below
Planner	Assignment plus live poses	Per-cone velocity command at 10 Hz	ORCA at $r = 0.60$ m; plain goal-seeking when no neighbour is within range
Fleet state	Telemetry stream	Health view, staleness flags, event log	Per-cone sequence numbers and last-heard timestamps

WHY BOTTLENECK ASSIGNMENT RATHER THAN THE USUAL HUNGARIAN MINIMUM-SUM

The layout is not ready until the *last* cone arrives (F3.6). Minimising the total distance travelled can leave one cone with a long haul while five sit idle; minimising the *maximum* single distance is what actually minimises reconfiguration time, which is what NFR-3 measures. Solve it by binary search over a distance threshold with a bipartite feasibility check at each step.

Implement minimum-sum first — in Python it is one call to `scipy.optimize.linear_sum_assignment` — then measure it, then show the bottleneck variant beating it on NFR-3. That comparison is a report figure on its own, and it costs an afternoon.

12.2 Cone firmware — state machine

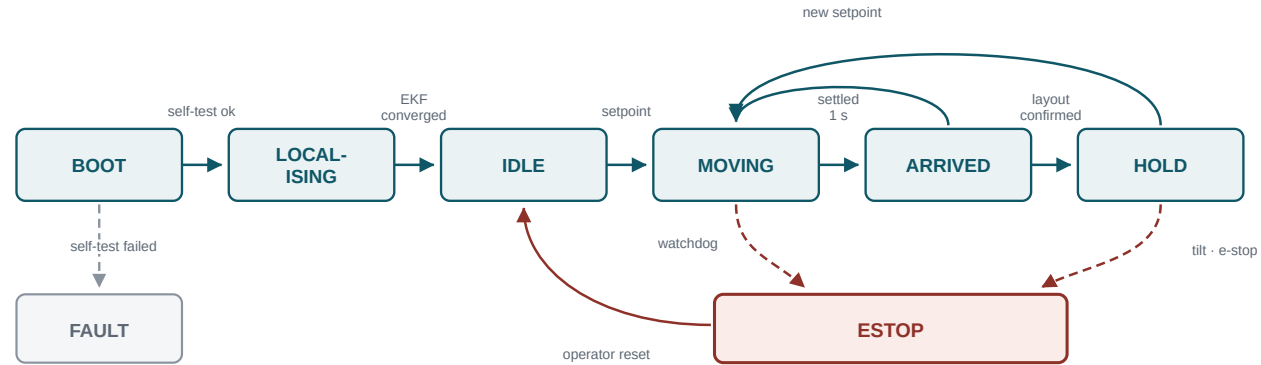


Figure 5. Cone state machine. Every transition into ESTOP cuts motor drive **in hardware**, not only in software. The watchdog transition is how NFR-9 is met without the coordinator having to be reachable: no valid command for 500 ms → ESTOP.

12.3 Control cascade on the cone

Three nested loops, each at its own rate.

LOOP	RATE	INPUT	OUTPUT
Wheel velocity PID ($\times 2$)	200 Hz	Encoder counts, via the ESP32 PCNT hardware counters	PWM duty
Unicycle pose controller	50 Hz	EKF pose and target pose	v, ω
Command and avoidance	10 Hz	ESP-NOW setpoint	Target pose or velocity

TWO IMPLEMENTATION NOTES THAT WILL OTHERWISE COST A WEEK EACH

Use the PCNT peripheral for quadrature decoding, not pin interrupts. At 1.0 m/s with a 12 CPR encoder behind a 30:1 gearbox, each wheel generates roughly 4 kHz of edges. Interrupt-driven counting will steal cycles from the 200 Hz control loop and the symptom will look like a tuning problem rather than a resourcing one.

Asphalt needs friction compensation. Add a static feed-forward term to the PID output so the wheel begins turning at low commands instead of stalling inside the deadband. Without it, terminal approach at 0.3 m/s (§11.4) will stick short of the target and NFR-1 will fail for reasons that have nothing to do with localisation.

13 Recommended technology stack

No stack preference was expressed, so this is a recommendation with its reasoning attached. Any part of it may be substituted — nothing in §11 or §12 depends on the choice of language.

LAYER	CHOICE	REASONING
Coordinator	Python 3.11+ — numpy, <code>scipy.optimize</code> , asyncio, websockets	Assignment is a library call and ORCA is a couple of hundred lines. Plotting for the report comes free, and iteration speed matters more here than runtime speed at 10 Hz
Cone firmware	C++ on ESP32-S3, Arduino core or ESP-IDF	Dual core lets the radio and the control loop stop competing; hardware PCNT and MCPWM peripherals are exactly what §12.3 needs; mature driver support for DW3000 and common IMUs
Tablet application	Web app — HTML Canvas, served by the coordinator	Runs on any tablet with a browser. No app store, no Android/iOS split, no installation: the instructor joins a hotspot and opens a URL. This is also what makes NFR-13 achievable
Simulator	Python, sharing the coordinator's planner code verbatim	See §14.1 — “verbatim” is the entire point

ON ROS 2

ROS 2 is the more academically impressive answer and it supplies Nav2, tf2, rviz and Gazebo at no cost. It is **not** recommended here, for two reasons. Its learning curve would consume a large fraction of a two-semester budget for a team that has not used it; and almost none of the heavy machinery it provides — SLAM, costmaps, sensor pipelines — is needed on a flat, empty, fully instrumented pad.

This decision flips entirely on one fact: if any student already knows ROS 2 well, revisit it immediately. The argument above is about the cost of learning it, not about its merits.

14 Development strategy

14.1 Build the simulator first

This is the most important process recommendation in this document.

Write the coordinator against a *cone interface*, then provide two implementations of it: the serial gateway, and a simulator that integrates differential-drive kinematics with injected noise — UWB jitter, wheel slip, packet loss, latency. The planner cannot tell them apart.

WHAT IT BUYS	WHY IT MATTERS HERE
Three of four roles never wait on the machine shop	The single most common way a four-person hardware capstone loses a month
Algorithms are debugged at twelve cones on a laptop	Then demonstrated on six real ones — and NFR-14 is satisfied as a by-product
Failures become repeatable tests	30% packet loss, a cone that stops answering, an anchor going down: all reproducible on demand, none of them reproducible on hardware
A demonstrable deliverable in its own right	If hardware slips, the simulator still shows the coordination working — the project degrades instead of failing

14.2 Build the measurement instrument early

The overhead-camera ground-truth rig (§11.2) and the F8.2 accuracy log are *prerequisites* for the evidence in §8, not by-products of it. Both should exist by week 8 of the first semester. From that point, every test run generates report material automatically and at no marginal cost.

14.3 Integrate in this order, and not out of it

One cone → two cones → six cones. Two cones is where collision avoidance first becomes real, and it is a far larger step than two to six. **Order parts for six, but build and prove two.**

15 Team split

Four students, four vertical slices. Each student owns a slice end to end — design, build, test, and the corresponding report chapter — so that no student's mark depends on another student's subsystem working.

ROLE	STUDENT	OWNS	FEATURES
Mechanical	MECH	Chassis and drivetrain geometry, wheel and motor selection, cone shell and impact strategy, centre-of-mass and tip-over analysis, weatherproofing, CAD and manufacture, battery access	NFR-11, NFR-12, NFR-15, F5.4 (mechanical)
Embedded & Power	MECA-1	Electronics integration and wiring or PCB, power system and BMS, motor drivers, firmware base, encoder and IMU drivers, wheel PID, self-test, firmware safety interlocks	F6.2, F6.3, F6.5, F6.6, F6.7, F5.2, F5.4, F5.5, F5.7, F5.8, NFR-8
Localisation & Comms	MECA-2	Anchor infrastructure and survey, UWB ranging, EKF pose fusion, ESP-NOW protocol and gateway, calibration routines, the ground-truth camera rig	F6.1, F7.1, F7.3, F5.3, NFR-1, NFR-2, NFR-6
Coordination & UI	MECA-3	Coordinator service, simulator, layout compiler, assignment, ORCA planner, tablet application, logging	F1.*, F2.*, F3.*, F4.*, F8.*, F5.1, F5.6, NFR-3, NFR-13, NFR-14

Shared, and scheduled explicitly rather than assumed: system integration, the test campaign of §8, the report, the presentation and the video.

FREEZE THESE TWO INTERFACES BY WEEK 6

These are where the four slices meet, and they are where integration projects usually fail:

- The cone command and telemetry message set — between MECA-1 and MECA-2/3.
- The coordinator ↔ cone abstraction (§14.1) — so MECA-3 can develop against the simulator while hardware is still on the bench.

After week 6, treat changes to either as change requests with all four students present, not as edits one person makes.

16 Timeline

Two semesters of roughly fifteen weeks. Milestones are stated as demonstrable outcomes, never as activities: "one cone drives to a commanded point" is a milestone, "work on localisation" is not.

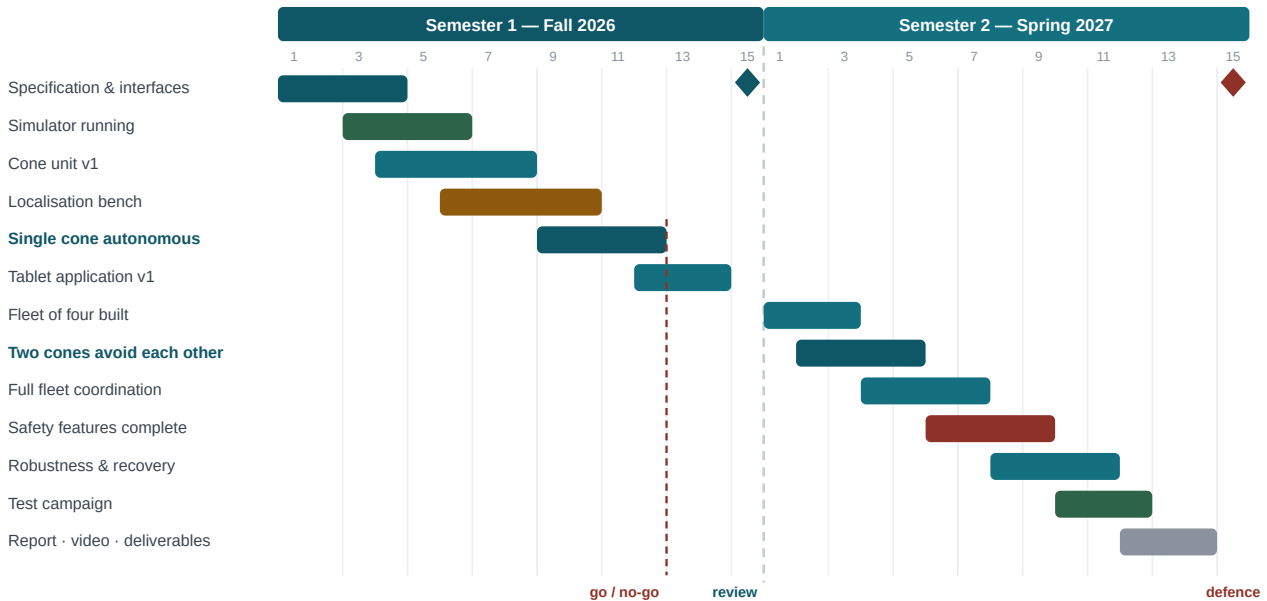


Figure 6. Two-semester schedule. Note the overlap in the first eight weeks: the simulator, the cone unit and the localisation bench proceed in parallel because none of them blocks another — that parallelism is what §14.1 exists to create.

16.1 Semester 1 — Fall 2026: prove the concept

WEEKS	MILESTONE	DONE WHEN
1–2	Specification frozen	All five questions in §10 answered by the advisor; fleet size, pad and heading requirement fixed
3–4	Interfaces frozen	The two interfaces in §15 documented and agreed by all four students
3–6	Simulator running	Six simulated cones move from a drawn path to targets, with assignment, ORCA, injected noise and packet loss. MECA-3 is now permanently unblocked
4–8	Cone unit v1	One chassis drives under manual command; encoders and PID closed; shell CAD released
6–10	Localisation bench	UWB ranging to four anchors, raw accuracy characterised, EKF fusing odometry and IMU
9–12	One cone drives to a commanded point autonomously	Position error measured against the camera rig. The semester's critical result
12–14	Tablet application v1	Draw, validate, preview and execute — against the simulator
15	Semester review	Live demonstration: one real cone driving to tablet-specified points, plus six in simulation

GO / NO-GO — WEEKS 9–12

This is the project's decision point. If a single cone cannot reach a commanded position within tolerance, the swarm work has nothing to stand on. **Escalate to the advisor immediately rather than proceeding to build more units** — the failure modes are cheap to fix at one unit and expensive at six. The most likely causes, in order, are localisation accuracy (§11.2), friction deadband (§12.3) and wheel slip (RSK-2).

16.2 Semester 2 — Spring 2027: make it a system

WEEKS	MILESTONE	DONE WHEN
1–3	Fleet of four built	Four units assembled from the v1 design, with shells fitted
2–5	Two cones avoid each other	Head-on and crossing conflicts resolved, on hardware. The first genuine swarm result
4–7	Full fleet coordination	Four cones deploy a preset layout; NFR-3 and NFR-7 measured
6–9	Safety features complete	F5.1 – F5.8 implemented and deliberately provoked; NFR-9 measured
8–11	Robustness	Tip detection and re-deploy (F3.8), battery management (F6.7), knock-over recovery per UC-3
10–12	Test campaign	The §8 verification matrix executed and the evidence collected
12–14	Deliverables	Report, presentation, subtitled video, source code, schematics, CAD
15	Defence	The four-minute sequence of §7.2, rehearsed

17 Sequencing rules

Three rules that matter more than any of the dates above.

#	RULE	RATIONALE
1	Simulator before hardware	Three of the four slices can progress without a physical cone. Any week in which MECA-3 is blocked on the machine shop is a planning failure, not a manufacturing one

#	RULE	RATIONALE
2	One, then two, then six	Two cones is where collision avoidance becomes real, and it is a much larger step than two to six. Do not build four units before two work
3	Build the measurement rig before you need the measurements	Ground truth and the F8.2 log should exist by week 8 of Semester 1. Every test run afterwards then generates report evidence for free (§14.2)

18 Risk register

ID	RISK	IMPACT	LIKELIHOOD	MITIGATION
RSK-1	Outdoor UWB accuracy worse than 25 cm	High NFR-1 fails	Medium	Characterise by week 10 of Semester 1, not later. Raise anchors on poles, improve anchor geometry, fuse harder. Fall back to an indoor demonstration with the camera — which §10 question 3 exists to decide <i>early</i> rather than under pressure
RSK-2	Wheel slip and traction on asphalt	Medium	Medium	Wheels ≥ 100 mm with rubber tread, torque margin $\geq 3\times$, closed-loop control, and UWB continuously correcting odometry drift
RSK-3	Budget will not cover six units	Medium	Medium	Design for six, build two, prove, then replicate. §7.2 needs only four
RSK-4	Integration compressed into the final weeks	High	High the classic capstone failure	The frozen interfaces of §15 and the simulator of §14.1 exist specifically to make integration continuous instead of terminal
RSK-5	Mechanical durability under vehicle contact	Medium	Medium	Resolve §10 question 5 early. Compliant sacrificial shell, mechanically isolated electronics, low centre of mass
RSK-6	One student's slice slips and blocks the others	High	Medium	Vertical slices plus the simulator mean each slice degrades independently rather than blocking the remainder
RSK-7	Scope creep from the swarm literature	Medium	High	§9 is the defence. Anything outside the Must set of §7.1 waits, however interesting it is

19 What to do first

In order, starting now:

1. **Take §10 to the advisor and close all five questions.** Questions 3 and 4 first — together they can remove most of the project's technical risk at no cost to its academic value.
2. **Fix the fleet size and the pad assumption,** and update §3 accordingly.
3. **Agree the two interfaces in §15 and write them down.**
4. **Start the simulator.** It is the longest pole with no dependencies, and everything in §14.1 follows from having it early.

APPENDICES

REFERENCE

A Feature index

All 53 features in one table, with the MoSCoW priority from §7.1 and the owning role from §15. This is the checklist against which progress is tracked.

ID	FEATURE	AREA	PRIORITY	OWNER
F1.1	Freehand path	Course authoring	MUST	MECA-3
F1.2	Point placement	Course authoring	SHOULD	MECA-3
F1.3	Geometric primitives	Course authoring	SHOULD	MECA-3
F1.4	Path editing	Course authoring	SHOULD	MECA-3
F1.5	Automatic spacing	Course authoring	MUST	MECA-3
F1.6	Numeric dimensioning	Course authoring	SHOULD	MECA-3
F1.7	Feasibility check	Course authoring	MUST	MECA-3
F1.8	Layout transform	Course authoring	SHOULD	MECA-3
F1.9	Reference overlay	Course authoring	COULD	MECA-3
F2.1	Save layout	Layout library	MUST	MECA-3
F2.2	Load layout	Layout library	MUST	MECA-3
F2.3	Standard presets	Layout library	MUST	MECA-3
F2.4	Manage layouts	Layout library	SHOULD	MECA-3
F2.5	Import / export	Layout library	COULD	MECA-3
F2.6	Lesson sequence	Layout library	COULD	MECA-3
F3.1	Preview with ETA	Deployment	MUST	MECA-3
F3.2	Execute	Deployment	MUST	MECA-3
F3.3	Live progress	Deployment	MUST	MECA-3
F3.4	Pause / resume	Deployment	SHOULD	MECA-3
F3.5	Abort	Deployment	SHOULD	MECA-3
F3.6	READY signal	Deployment	MUST	MECA-3
F3.7	Gather	Deployment	SHOULD	MECA-3
F3.8	Re-deploy displaced	Deployment	SHOULD	MECA-3
F4.1	Live map	Monitoring	MUST	MECA-3
F4.2	Per-cone status	Monitoring	MUST	MECA-3
F4.3	Fleet summary	Monitoring	SHOULD	MECA-3
F4.4	Alerts	Monitoring	SHOULD	MECA-3
F4.5	Event log	Monitoring	SHOULD	MECA-3
F5.1	Emergency stop	Safety	MUST	MECA-3
F5.2	Comms watchdog	Safety	MUST	MECA-1
F5.3	Geofence	Safety	MUST	MECA-2
F5.4	Tip-over detection	Safety	MUST	MECA-1 · MECH
F5.5	Obstacle reflex	Safety	SHOULD	MECA-1
F5.6	Vehicle interlock	Safety	SHOULD	MECA-3
F5.7	Speed limiting	Safety	MUST	MECA-1
F5.8	Movement conspicuity	Safety	SHOULD	MECA-1
F6.1	Self-localisation	Cone autonomy	MUST	MECA-2
F6.2	Drive to pose	Cone autonomy	MUST	MECA-1
F6.3	Hold position	Cone autonomy	MUST	MECA-1
F6.4	Mutual avoidance	Cone autonomy	MUST	MECA-3
F6.5	Telemetry	Cone autonomy	MUST	MECA-1
F6.6	Power-on self-test	Cone autonomy	MUST	MECA-1
F6.7	Battery management	Cone autonomy	SHOULD	MECA-1
F6.8	Identify	Cone autonomy	SHOULD	MECA-1
F7.1	Pad definition	Administration	MUST	MECA-2
F7.2	Fleet registration	Administration	MUST	MECA-3
F7.3	Calibration	Administration	MUST	MECA-2
F7.4	Firmware update	Administration	COULD	MECA-1

ID	FEATURE	AREA	PRIORITY	OWNER
F7.5	Diagnostics	Administration	SHOULD	MECA-1
F7.6	Defaults	Administration	COULD	MECA-3
F8.1	Session record	Data	SHOULD	MECA-3
F8.2	Accuracy record	Data	SHOULD	MECA-3
F8.3	Export	Data	COULD	MECA-3

Totals: 53 features — MUST 26 SHOULD 21 COULD 6. Every feature in the Safety area is mandatory regardless of the priority shown; see §2.5.

B Derived parameters and design values

Every number the design depends on, in one place, with the section that derives it. When one of these changes, check the source section for what else moves with it.

PARAMETER	VALUE	SOURCE
MOTION		
Maximum coordinated speed, v_{max}	0.80 m/s	§11.4, F5.7
Terminal approach speed, within 1 m of goal	0.30 m/s	§11.4
Cone physical radius, r_{cone}	0.20 m	§11.4
ORCA safety radius, r_{safety}	0.60 m	§11.4 — derived, not assumed
Worst-case control staleness	190 ms	§11.4
Settling dwell before ARRIVED	1 s	§12.2
RATES		
Coordinator planning cycle	10 Hz — 100 ms	§12.1
Telemetry reporting rate	10 Hz	F6.5
Pose controller loop	50 Hz	§12.3
Wheel velocity PID loop	200 Hz	§12.3
UWB fix rate — 6 cones / 10 cones	≈ 13 Hz / ≈ 8 Hz	§11.2
Live map refresh	≥ 5 Hz	NFR-5
SAFETY		
Communications watchdog timeout	500 ms	F5.2, NFR-9
Obstacle reflex brake threshold	0.40 m	F5.5
Tip detection latency	< 1 s	F5.4
E-STOP to fleet halt	≤ 500 ms	NFR-9
ACCURACY AND TOLERANCE		
Position error, RMS / 95th percentile	0.15 m / 0.25 m	NFR-1
Heading error	10°	NFR-2
Reconfiguration budget, 6 cones over 20 × 20 m	60 s	NFR-3
Command to first movement	≤ 500 ms	NFR-4
Command link packet loss at 50 m	< 2%	NFR-6
LAYOUT AND FLEET		
Default cone spacing along a path	2.0 m	F1.5
Spacing adjustment range	0.5 – 20 m	F1.5
Prototype fleet baseline	6 cones	§3 — open question 1
Fleet verified in simulation	12 cones	NFR-14
Minimum demonstrable product fleet	4 cones	§7.2
INFRASTRUCTURE AND ENVIRONMENT		
UWB anchor count	4	§11.2
UWB anchor mounting height	3 – 4 m	§11.2 — multipath mitigation
Assumed pad dimensions	30 × 50 m	§1.5 — open question 2
Maximum gradient	5%	§1.5
Operating temperature	0 – 45 °C	NFR-10
Ingress protection	IP54 minimum	NFR-12
POWER AND INTERFACE		
Battery warning / lockout thresholds	20% / 10%	F6.7
Endurance per charge	≥ 2 h	NFR-8
Battery replacement time	< 5 min, hand tools	NFR-15
Minimum touch target	48 px	§5.1

C Glossary

TERM	MEANING IN THIS DOCUMENT
ArUco	A square fiducial marker readable by a camera, giving full pose. Used here only on the ground-truth rig (§11.2)
BMS	Battery management system — cell balancing and over-discharge protection
Bottleneck assignment	Cone-to-target matching that minimises the <i>largest</i> single distance rather than the total. Chosen because the layout is ready only when the last cone arrives (§12.1)
EKF	Extended Kalman Filter — fuses UWB position with wheel odometry and gyroscope into a single pose estimate
ESP-NOW	Espressif's connectionless 2.4 GHz link. No access point, 2–5 ms latency, 250-byte payload. The control link (§11.3)
GDOP	Geometric dilution of precision — how much poor anchor geometry amplifies ranging error into position error
Geofence	A boundary a cone refuses to cross regardless of command (F5.3)
Holonomic	Able to move instantaneously in any direction. Differential-drive cones are <i>not</i> , which is why ORCA output must be wrapped by a unicycle controller (§11.4)
Hungarian algorithm	Classical $O(n^3)$ solution to minimum-sum assignment. Implemented first as a baseline, then compared against the bottleneck variant (§12.1)
IMU	Inertial measurement unit — accelerometer and gyroscope. Supplies heading rate and tip detection
ISO 3888-2	The standard double-lane-change ("moose test") course geometry, used as a preset (§3)
MoSCoW	Must / Should / Could / Won't prioritisation (§7.1). It measures build sequencing, never permission to omit a safety feature
ORCA	Optimal Reciprocal Collision Avoidance — each agent takes half the responsibility for avoiding each neighbour. The chosen avoidance method (§11.4)
PCNT	The ESP32's hardware pulse-counter peripheral. Used for quadrature decoding so encoder edges do not consume CPU (§12.3)
PID	Proportional–integral–derivative controller. Here, the innermost wheel-velocity loop at 200 Hz
Pose	Position and heading together — (x, y, θ)
RMS	Root mean square. NFR-1 is stated in RMS because a single worst case is not representative and a mean hides the tail; the 95th percentile bounds the tail separately
RTK-GPS	Satellite positioning corrected by a fixed base station, giving ~2 cm. Rejected on cost per unit (§11.2)
TDMA	Time-division multiple access — cones take turns ranging, which is why the UWB fix rate falls as the fleet grows (§11.2)
TDoA	Time difference of arrival — tags transmit and synchronised anchors listen. The scaling path beyond ~10 cones (§11.2)
TWR	Two-way ranging — the baseline UWB method: a tag and an anchor exchange packets and derive distance from the round trip
Unicycle model	Describing a differential-drive robot by forward speed v and turn rate ω rather than by two wheel speeds
UWB	Ultra-wideband radio. Used for ranging, not data: 10–30 cm accuracy over ~100 m, and the chosen basis for localisation (§11.2)